



IT-SIMPLERA INSTITUTE

Network Administration Internship Program

WEEK 07 TASK REPORT

Enterprise VPN, Firewall, and Secure Access Implementation

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1. Assignment Overview

This report documents the design, implementation, and verification of an enterprise-grade secure network for Week 7 of the Network Administration Internship Program at IT-Simplera Institute. The objective of this assignment was to build a two-site enterprise network a Headquarters (HQ) site and a Branch site and secure it end-to-end using a combination of VPN, firewall, authentication, routing security, and monitoring technologies.

The scope of the assignment covered the following technology areas:

- IPsec Site-to-Site VPN for encrypted branch-to-branch communication
- Zone-Based Firewall (ZBF) for policy-based traffic inspection between security zones
- AAA authentication and SSH version 2 for secure device management
- OSPF routing with MD5 authentication, passive interfaces, and prefix-list filtering
- Centralized monitoring using Syslog, NTP, and SNMPv3
- Basic device hardening to reduce the network's attack surface

All configuration and testing was performed in GNS3, using Cisco c7200 routers and Cisco IOSvL2 switches, hosted on a GNS3 server running inside an Ubuntu virtual machine.

2. Network Topology

The topology consists of two sites connected over a simulated WAN link. R1 represents the HQ router and R2 represents the Branch router. Each site has its own LAN switch and end-user PC, and the HQ site additionally hosts a Docker-based Syslog server used for centralized logging.

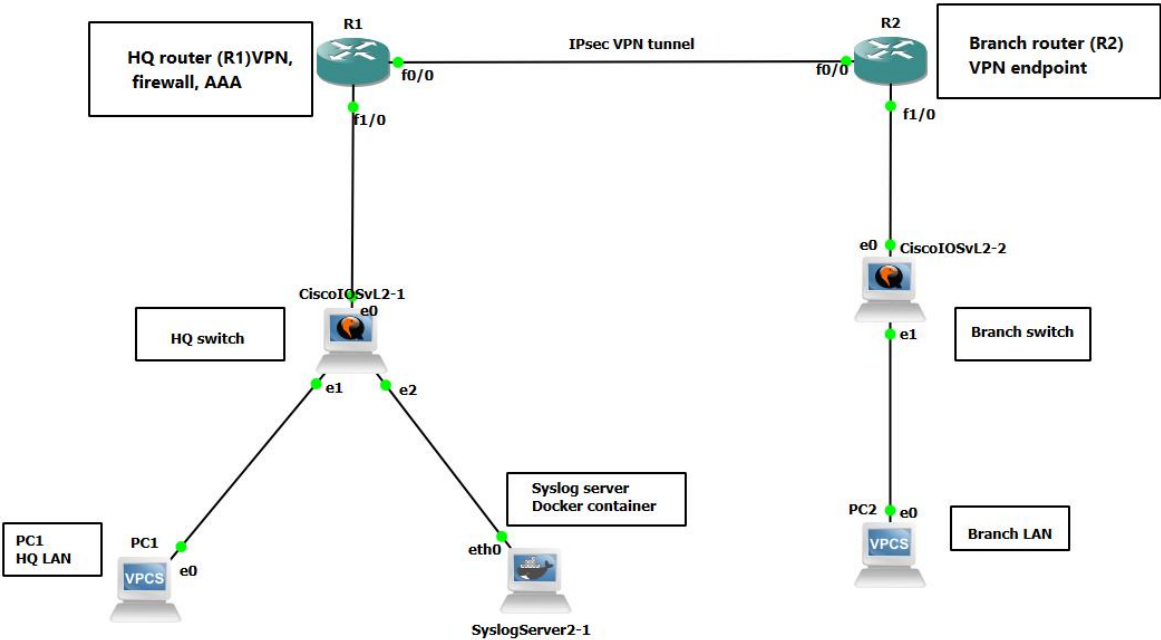


Figure 2.1 — Network Topology

2.1 IP Addressing Scheme

The table below summarizes the IP addressing plan used across the topology. The WAN link between R1 and R2 uses a /30 point-to-point subnet, while each site's LAN uses a /24 subnet.

Device	Interface	IP Address	Subnet	Role
R1 (HQ Router)	FastEthernet0/0	203.0.113.1	255.255.255.252	WAN link to R2
R1 (HQ Router)	FastEthernet1/0	192.168.10.1	255.255.255.0	HQ LAN gateway
R2 (Branch Router)	FastEthernet0/0	203.0.113.2	255.255.255.252	WAN link to R1
R2 (Branch Router)	FastEthernet1/0	192.168.20.1	255.255.255.0	Branch LAN gateway
PC1	e0	192.168.10.10	255.255.255.0	HQ LAN host

Syslog Server	eth0	192.168.10.20	255.255.255.0	HQ LAN — Docker (Alpine, busybox syslogd)
PC2	e0	192.168.20.10	255.255.255.0	Branch LAN host

3. Security Architecture

The security design follows a defence-in-depth approach, layering multiple independent controls so that no single misconfiguration compromises the whole network:

- **Perimeter encryption:** An IPsec Site-to-Site VPN tunnel encrypts all traffic between the HQ and Branch LANs as it crosses the WAN link, protecting confidentiality and integrity even though the underlying link is untrusted.
- **Traffic policy enforcement:** A Zone-Based Firewall on R1 places the LAN interface in an INSIDE zone and the WAN interface in an OUTSIDE zone, and only permits and inspects traffic that originates from INSIDE protecting the internal network from unsolicited external connections.
- **Access control to devices:** AAA with a local user database, combined with SSH version 2, ensures that only authenticated administrators can manage the routers, and that management sessions themselves are encrypted.
- **Routing integrity:** OSPF MD5 authentication prevents a rogue device from injecting false routes into the network, while passive interfaces stop routing advertisements from leaking onto end-user LANs, and prefix-list filtering restricts which routes are accepted.
- **Visibility:** Syslog, NTP, and SNMPv3 together provide a centralized, time-synchronized, and securely-monitored view of what is happening across the network, which is essential for detecting and investigating incidents.
- **Reduced attack surface:** Basic hardening (disabling CDP, disabling the HTTP server, shutting down unused interfaces, and enabling password encryption) removes unnecessary services and entry points.

4. Configuration Steps

This section documents, technology by technology, what was configured, why it was necessary, and the exact commands used.

4.1 Basic IP Addressing

Before any security feature could be layered on, the routers needed reachable interfaces. R1 and R2 were each given a WAN-facing interface (on the 203.0.113.0/30 subnet) and a LAN-facing interface acting as the default gateway for their respective site.

```
! R1
interface f0/0
ip address 203.0.113.1 255.255.255.252
no shutdown
interface f1/0
ip address 192.168.10.1 255.255.255.0
no shutdown

! R2 (mirrored)
interface f0/0
ip address 203.0.113.2 255.255.255.252
no shutdown
interface f1/0
ip address 192.168.20.1 255.255.255.0
no shutdown
```

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          203.0.113.1     YES manual up          up
FastEthernet1/0          192.168.10.1    YES manual up          up
Serial2/0                 unassigned      YES unset  administratively down down
Serial2/1                 unassigned      YES unset  administratively down down
Serial2/2                 unassigned      YES unset  administratively down down
Serial2/3                 unassigned      YES unset  administratively down down
```

Figure 4.1.1 — R1 interface verification

```
R2#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          203.0.113.2     YES manual up          up
FastEthernet1/0          192.168.20.1    YES manual up          up
Serial2/0                 unassigned      YES unset  administratively down down
Serial2/1                 unassigned      YES unset  administratively down down
Serial2/2                 unassigned      YES unset  administratively down down
Serial2/3                 unassigned      YES unset  administratively down down
R2#
```

Figure 4.1.2 — R2 interface verification

```
R1#ping 203.0.113.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 28/37/60 ms
```

Figure 4.1.3 — WAN link connectivity test

4.2 Syslog Configuration

Syslog centralizes log messages from network devices onto a dedicated server, which is essential for auditing and troubleshooting. A lightweight Syslog server was created as a Docker container (Alpine Linux, running BusyBox's syslogd) inside the GNS3 topology on the HQ LAN, and R1 was configured to forward its log messages to it.

```
! Syslog server container
ip addr add 192.168.10.20/24 dev eth0
ip route add default via 192.168.10.1
busybox syslogd -n
! R1
logging host 192.168.10.20
logging trap informational
logging on
```

```
/ # ip addr show eth0
8: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN
qlen 1000
    link/ether 02:42:4c:c8:e4:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.20/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::42:4cff:fec8:e400/64 scope link
        valid_lft forever preferred_lft forever
/ #
```

Figure 4.2.1 — Syslog server IP setup

```
R1#ping 192.168.10.20
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.20, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 16/39/80 ms
R1#
```


Figure 4.2.2 — R1 to Syslog server connectivity

```
*Aug 15 08:19:20.151: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.10.20 po
rt 514 CLI Request Triggered
*Aug 15 08:19:21.155: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.10.20 po
rt 514 started - CLI initiated
*Aug 15 08:19:37.639: %SYS-5-CONFIG_I: Configured from console by console
```

Figure 4.2.3 — Syslog forwarding confirmation

```
Trap logging: level informational, 56 message lines logged
Logging to 192.168.10.20 (udp port 514, audit disabled,
link up),
17 message lines logged,
0 message lines rate-limited,
0 message lines dropped-by-MD,
xml disabled, sequence number disabled
filtering disabled
Logging Source-Interface:      VRF Name:
Log Buffer (8192 bytes):
```

Figure 4.2.4 — Syslog verification summary

4.3 NTP Configuration

Network Time Protocol (NTP) keeps all devices' clocks synchronized, which is critical for correlating log timestamps across the network. R1 was configured as an NTP master (stratum 3), and R2 was configured as its NTP client.

```
! R1 (NTP master)
ntp master 3

! R2 (NTP client)
ntp server 203.0.113.1
```

```
R1#show ntp status
Clock is synchronized, stratum 3, reference is 127.127.1.1
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**19
ntp uptime is 3800 (1/100 of seconds), resolution is 4000
reference time is EE2AA5D0.7EDAE97B (08:46:08.495 UTC Sat Aug 15 2026)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 1937.71 msec, peer dispersion is 1937.62 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
system poll interval is 16, last update was 6 sec ago.
R1#
```

Figure 4.3.1 — R1 NTP status

```
R2#show ntp status
Clock is synchronized, stratum 4, reference is 203.0.113.1
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**19
ntp uptime is 12400 (1/100 of seconds), resolution is 4000
reference time is EE2AA60C.0DB56CAA (08:47:08.053 UTC Sat Aug 15 2026)
clock offset is 21.7787 msec, root delay is 67.88 msec
root dispersion is 4212.38 msec, peer dispersion is 187.57 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
system poll interval is 64, last update was 54 sec ago.
R2#
```

Figure 4.3.2 — R2 NTP status

4.4 SNMPv3 Configuration

SNMP version 3 was used instead of v1/v2c because it adds both authentication and encryption to management traffic, which the assignment specifically required. A group and user were created on both routers with SHA authentication and AES-128 encryption (privacy), and traps were pointed at the Syslog server host.

```
snmp-server group ADMIN v3 priv
snmp-server user netadmin ADMIN v3 auth sha AuthPass123 priv aes 128 PrivPass123
snmp-server enable traps
snmp-server host 192.168.10.20 version 3 priv netadmin
```

```
R2#show snmp user

User name: netadmin
Engine ID: 800000090300CA027B540000
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: ADMIN

R2#show snmp group
groupname: ILMI                security model:v1
contextname: <no context specified> storage-type: permanent
readview : *ilmi              writeview: *ilmi

notifyview: <no notifyview specified>
row status: active

groupname: ILMI                security model:v2c
contextname: <no context specified> storage-type: permanent
readview : *ilmi              writeview: *ilmi

notifyview: <no notifyview specified>
row status: active

groupname: ADMIN              security model:v3 priv
contextname: <no context specified> storage-type: nonvolatile
readview : v1default          writeview: <no writeview specified>

notifyview: <no notifyview specified>
row status: active
```

Figure 4.4.1 — R1 SNMPv3 user

```
R1#show snmp user

User name: netadmin
Engine ID: 800000090300CA012F900000
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: ADMIN
```

```

R1#show snmp group
groupname: ILMI                                security model:v1
contextname: <no context specified>            storage-type: permanent
readview : *ilmi                               writeview: *ilmi

notifyview: <no notifyview specified>
row status: active

groupname: ILMI                                security model:v2c
contextname: <no context specified>            storage-type: permanent
readview : *ilmi                               writeview: *ilmi

notifyview: <no notifyview specified>
row status: active

groupname: ADMIN                               security model:v3 priv
contextname: <no context specified>            storage-type: nonvolatile
readview : v1default                           writeview: <no writeview specified>

notifyview: <no notifyview specified>
row status: active

```

Figure 4.4.2 — R2 SNMPv3 user and group

4.5 IPsec Site-to-Site VPN

An IPsec Site-to-Site VPN tunnel was built between R1 and R2 to encrypt all traffic exchanged between the HQ and Branch LANs across the WAN link. The configuration follows the standard two-phase IPsec model:

- Phase 1 (ISAKMP/IKE): Establishes a secure, authenticated management channel between the two routers using AES-256 encryption, SHA-256 hashing, Diffie-Hellman group 14, and pre-shared key authentication.
- Phase 2 (IPsec): Builds the actual data-plane tunnel using a transform-set (ESP with AES-256 encryption and SHA-256-HMAC authentication) and a crypto map that ties the peer, transform-set, and an access-list defining 'interesting traffic' together.

```

! Phase 1 — ISAKMP policy (R1, mirrored on R2 with peer reversed)
crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
 lifetime 3600
crypto isakmp key VpnSecret123 address 203.0.113.2

! Phase 2 — interesting traffic, transform-set, crypto map
access-list 100 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
crypto ipsec transform-set TSET esp-aes 256 esp-sha256-hmac
 mode tunnel
crypto map VPNMAP 10 ipsec-isakmp
 set peer 203.0.113.2
 set transform-set TSET
 match address 100

! Apply to the WAN interface
interface f0/0
 crypto map VPNMAP

! Static routes so LAN-to-LAN traffic reaches the crypto map
ip route 192.168.20.0 255.255.255.0 203.0.113.2 (on R1)
ip route 192.168.10.0 255.255.255.0 203.0.113.1 (on R2)

```

R2's configuration mirrors R1's exactly, with the peer address and the access-list source/destination reversed.


```

R1#show crypto map
Crypto Map IPv4 "VPNMAP" 10 ipsec-isakmp
  Peer = 203.0.113.2
  Extended IP access list 100
    access-list 100 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
  Current peer: 203.0.113.2
  Security association lifetime: 4608000 kilobytes/3600 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): N
  Mixed-mode : Disabled
  Transform sets={
    TSET: { esp-256-aes esp-sha256-hmac } ,
  }
  Interfaces using crypto map VPNMAP:
    FastEthernet0/0

```

Figure 4.5.1 — R1 crypto map

```

R2#show crypto map
Crypto Map IPv4 "VPNMAP" 10 ipsec-isakmp
  Peer = 203.0.113.1
  Extended IP access list 100
    access-list 100 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
  Current peer: 203.0.113.1
  Security association lifetime: 4608000 kilobytes/3600 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): N
  Mixed-mode : Disabled
  Transform sets={
    TSET: { esp-256-aes esp-sha256-hmac } ,
  }
  Interfaces using crypto map VPNMAP:
    FastEthernet0/0
R2#

```

Figure 4.5.2 — R2 crypto map

```

R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
203.0.113.2  203.0.113.1  QM_IDLE        1001  ACTIVE

IPv6 Crypto ISAKMP SA

```

Figure 4.5.3 — ISAKMP Security Association


```

R1#show crypto ipsec sa

interface: FastEthernet0/0
  Crypto map tag: VPNMAP, local addr 203.0.113.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
  current_peer 203.0.113.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 4, #pkts encrypt: 4, #pkts digest: 4
    #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 4

inbound esp sas:
  spi: 0xD410F7B4(3557881780)
    transform: esp-256-aes esp-sha256-hmac ,
    in use settings = {Tunnel, }
    conn id: 1, flow_id: SW:1, sibling_flags 80004040, crypto map: VPNMAP
    sa timing: remaining key lifetime (k/sec): (4294648/3492)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0xCA85C34(3405274164)
    transform: esp-256-aes esp-sha256-hmac ,
    in use settings = {Tunnel, }
    conn id: 2, flow_id: SW:2, sibling_flags 80004040, crypto map: VPNMAP
    sa timing: remaining key lifetime (k/sec): (4294648/3492)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

outbound ah sas:

outbound pcp sas:

```

Figure 4.5.4 — IPsec Security Association and encrypted packet counters

4.6 Zone-Based Firewall (ZBF)

A Zone-Based Firewall was configured on R1 to control traffic flow between the internal LAN and the outside world. The LAN-facing interface was placed in an INSIDE zone and the WAN-facing interface in an OUTSIDE zone. A class-map identifies TCP, UDP, and ICMP traffic, and a policy-map inspects (stateful firewall) that traffic when it flows from INSIDE to OUTSIDE, while dropping everything else by default.

```

zone security INSIDE
zone security OUTSIDE

interface f1/0
  zone-member security INSIDE
interface f0/0
  zone-member security OUTSIDE

class-map type inspect match-any INSIDE-TO-OUTSIDE-CLASS
  match protocol tcp
  match protocol udp
  match protocol icmp

policy-map type inspect INSIDE-TO-OUTSIDE-POLICY
  class type inspect INSIDE-TO-OUTSIDE-CLASS
    inspect
  class class-default

```

```
drop
```

```
zone-pair security IN-TO-OUT source INSIDE destination OUTSIDE  
service-policy type inspect INSIDE-TO-OUTSIDE-POLICY
```

Because the policy uses stateful inspect (not a plain permit), return traffic for connections initiated from INSIDE is automatically allowed back in, while unsolicited traffic originating from OUTSIDE is dropped by the class-default rule.

```
R1#show zone security  
zone self  
  Description: System defined zone  
  
zone INSIDE  
  Member Interfaces:  
    FastEthernet1/0  
  
zone OUTSIDE  
  Member Interfaces:  
    FastEthernet0/0  
  
R1#show zone-pair security  
Zone-pair name IN-TO-OUT  
  Source-Zone INSIDE  Destination-Zone OUTSIDE  
  service-policy INSIDE-TO-OUTSIDE-POLICY  
  
R1#show zone-pair security  
Zone-pair name IN-TO-OUT  
  Source-Zone INSIDE  Destination-Zone OUTSIDE  
  service-policy INSIDE-TO-OUTSIDE-POLICY
```

Figure 4.6.1 — Security zones and their member interfaces

```
R1#show zone-pair security  
Zone-pair name IN-TO-OUT  
  Source-Zone INSIDE  Destination-Zone OUTSIDE  
  service-policy INSIDE-TO-OUTSIDE-POLICY
```

Figure 4.6.2 — Zone-pair and attached policy

```

R1#show policy-map type inspect zone-pair IN-TO-OUT

policy exists on zp IN-TO-OUT
Zone-pair: IN-TO-OUT

Service-policy inspect : INSIDE-TO-OUTSIDE-POLICY

Class-map: INSIDE-TO-OUTSIDE-CLASS (match-any)
  Match: protocol tcp
    0 packets, 0 bytes
    30 second rate 0 bps
  Match: protocol udp
    0 packets, 0 bytes
    30 second rate 0 bps
  Match: protocol icmp
    0 packets, 0 bytes
    30 second rate 0 bps

Inspect
  Session creations since subsystem startup or last reset 0
  Current session counts (estab/half-open/terminating) [0:0:0]
  Maxever session counts (estab/half-open/terminating) [0:0:0]
  Last session created never
  Last statistic reset never
  Last session creation rate 0
  Maxever session creation rate 0
  Last half-open session total 0
  TCP reassembly statistics
    received 0 packets out-of-order; dropped 0
    peak memory usage 0 KB; current usage: 0 KB
    peak queue length 0

Class-map: class-default (match-any)
  Match: any
  Drop
    0 packets, 0 bytes

```

Figure 4.6.3 — Firewall inspection counters confirming policy is active

4.7 AAA Authentication and SSH

To secure device management, a local AAA (Authentication, Authorization, and Accounting) database was configured on both routers, and Telnet was replaced with SSH version 2 for all remote access.

```

ip domain-name itsimplera.local
username netadmin privilege 15 secret Admin@12345
aaa new-model
aaa authentication login default local
aaa authentication enable default enable
crypto key generate rsa modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3

line vty 0 4
  transport input ssh
line console 0
  exec-timeout 5 0

banner motd #
AUTHORIZED ACCESS ONLY - IT-Simplera Network
#

```

Once `aaa new-model` is enabled, the router automatically applies the default authentication method-list to all lines (console and VTY), so no separate `'login local'` command is required on the line itself.

```
R1#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication timeout: 60 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDMCK15TlekD2VpYvBjeyhQ4HdRWbVjAkMDXLYhk+tU
rgXSVNJ/erigQZ/SH1clUgJg2Tb2QvTN17PqzKWpaIx+/pJi+3QHBXIVZojj30TGJvOS3u80r/5c+oC
S+7GCxCad9DnhIULsGS/E9KOE3vJUd9iIzkd+OJFguMyt7CfAYowxrKaG4MRBpKoEsaUt8qcc40qxTkA
jY/1YmeqXDWKEup8valJHvzBq/W+ky9/bB+fRaRUeYbw6KpNfsq2NqFNjpBErU6cRan0+MpXl6qtYanP
7XIWiFbMXMUCwTv4XyRqmVvNlsxd8yydejARIipfUfb3Hc+7FYg19KfEX5JN
```

Figure 4.7.1 — SSH status

```
R1#show running-config | section aaa
aaa new-model
aaa authentication login default local
aaa authentication enable default enable
aaa session-id common
```

Figure 4.7.2 — AAA configuration

```
R2#ssh -l netadmin 203.0.113.1
Password:

AUTHORIZED ACCESS ONLY - IT-Simplera Network

R1>
```

Figure 4.7.3 — Router-to-router SSH login, proving AAA + SSH work end-to-end

4.8 OSPF Authentication and Routing Security

OSPF was enabled between R1 and R2 to dynamically advertise the WAN and LAN networks, secured with MD5 authentication so that only routers presenting the correct key can form an adjacency. Passive-interface was applied on each router's LAN-facing interface so OSPF hello packets are not sent onto end-user LANs, and a prefix-list was used to filter which routes each router accepts from its neighbor.

```
! R1
router ospf 1
network 203.0.113.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
passive-interface FastEthernet1/0
area 0 authentication message-digest

interface f0/0
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 OspfSecret123

ip prefix-list FILTER-IN seq 5 permit 192.168.20.0/24
ip prefix-list FILTER-IN seq 10 deny 0.0.0.0/0 le 32
router ospf 1
distribute-list prefix FILTER-IN in

! R2 mirrors this configuration with its own LAN network and a
! prefix-list permitting 192.168.10.0/24 instead
```

```
R1#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
203.0.113.2      1     FULL/DR         00:00:32   203.0.113.2   FastEthernet0/0
```


Figure 4.8.1 — OSPF neighbor adjacency

```
R1#show ip ospf interface f0/0
FastEthernet0/0 is up, line protocol is up
Internet Address 203.0.113.1/30, Area 0, Attached via Network Statement
Process ID 1, Router ID 203.0.113.1, Network Type BROADCAST, Cost: 1
Topology-MTID      Cost      Disabled      Shutdown      Topology Name
      0          1          no            no            Base
Transmit Delay is 1 sec, State BDR, Priority 1
Designated Router (ID) 203.0.113.2, Interface address 203.0.113.2
Backup Designated router (ID) 203.0.113.1, Interface address 203.0.113.1
Flush timer for old DR LSA due in 00:01:31
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  oob-resync timeout 40
  Hello due in 00:00:06
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 203.0.113.2 (Designated Router)
Suppress hello for 0 neighbor(s)
Message digest authentication enabled
Youngest key id is 1
```

Figure 4.8.2 — OSPF MD5 authentication verification

```
R1#show ip prefix-list
ip prefix-list FILTER-IN: 2 entries
  seq 5 permit 192.168.20.0/24
  seq 10 deny 0.0.0.0/0 le 32
R1#
```

Figure 4.8.3 — Prefix-list route filtering

4.9 Basic Hardening

Finally, both routers were hardened to reduce the number of ways they could be probed or attacked: unnecessary services were disabled, all passwords in the configuration were encrypted, and unused Serial interfaces were administratively shut down.

```
no ip http server
no ip http secure-server
no cdp run
service password-encryption

interface Serial2/0
shutdown
interface Serial2/1
shutdown
interface Serial2/2
shutdown
interface Serial2/3
shutdown
```

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          203.0.113.1    YES manual up            up
FastEthernet1/0          192.168.10.1   YES manual up            up
Serial2/0                 unassigned      YES unset  administratively down down
Serial2/1                 unassigned      YES unset  administratively down down
Serial2/2                 unassigned      YES unset  administratively down down
Serial2/3                 unassigned      YES unset  administratively down down
R1#
```

Figure 4.9.1 — R1 hardening verification

```
R2#show running-config | include service password-encryption
service password-encryption
R2#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          203.0.113.2    YES manual up            up
FastEthernet1/0          192.168.20.1   YES manual up            up
Serial2/0                 unassigned      YES unset  administratively down down
Serial2/1                 unassigned      YES unset  administratively down down
Serial2/2                 unassigned      YES unset  administratively down down
Serial2/3                 unassigned      YES unset  administratively down down
R2#
```

Figure 4.9.2 — R2 hardening verification

5. Connectivity and Security Validation

After all security layers were configured, end-to-end connectivity was validated from the HQ LAN through the VPN tunnel and the firewall to the Branch LAN.

```
PC1> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=125.117 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=30.447 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=27.748 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=20.917 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=23.318 ms

PC1> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=254 time=116.234 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=254 time=53.011 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=254 time=50.837 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=254 time=51.486 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=254 time=43.366 ms

PC1>
```

Figure 5.1 — End-to-end connectivity validation

6. Troubleshooting Report

Several issues were encountered during implementation. Each is documented below along with the verification command used to diagnose it and the resolution applied.

6.1 HQ Switch Losing MAC-Learning Ability

Issue: After the GNS3 VM was put to sleep and resumed, the HQ-side IOSvL2 switch repeatedly showed an empty MAC address table (show mac address-table) even though its ports displayed a 'connected' status. This blocked all Layer 2 connectivity on the HQ LAN, including the syslog server and PC1.

Diagnosis: show interfaces status confirmed the physical ports were connected, and show mac address-table confirmed no MAC addresses were being learned — isolating the fault to the switch's internal forwarding state rather than the cabling or PC configuration. Bouncing the affected ports (shutdown / no shutdown) did not resolve it.

Resolution: The switch node was deleted from the topology and replaced with a freshly-instantiated IOSvL2 node, with the same connections rebuilt. This immediately restored normal MAC learning and connectivity.

6.2 Duplex Mismatch Between R1 and the HQ Switch

Issue: Even after replacing the switch, R1 could not reach devices on its own LAN. show interfaces f1/0 showed R1's interface as up/up with clean counters, but no MAC address for R1 ever appeared in the switch's table.

Diagnosis: show running-config interface f1/0 revealed the interface was set to duplex half, while the switch port reported a-full (auto full-duplex) a duplex mismatch that silently drops frames rather than bringing the link down.

Resolution: The interface was explicitly set to duplex full to match the switch, since this platform's interface did not support the duplex auto keyword. Connectivity was restored immediately after the change.

6.3 Missing Route for VPN-Protected Traffic

Issue: After configuring ISAKMP policy, transform-set, and crypto map on both routers, the IPsec tunnel failed to establish. show crypto isakmp sa returned an empty table, and debug crypto isakmp produced no output at all when interesting traffic was generated.

Diagnosis: show ip route on R1 showed no route to the Branch LAN (192.168.20.0/24) — traffic destined for the Branch network had no path out of the router at all, so it never reached the crypto map to trigger IKE negotiation.

Resolution: Static routes were added on both routers (ip route 192.168.20.0 255.255.255.0 203.0.113.2 on R1, and the mirror on R2) pointing across the WAN link. Once routing was in place, the very next ping successfully triggered a full IKE Phase 1 and Phase 2 negotiation, confirmed via debug crypto isakmp output showing the state transitioning to IKE_QM_PHASE2_COMPLETE.

6.4 Syslog Container Image Failure

Issue: The first Syslog server container, built from the rsyslog/rsyslog Docker image, exited immediately after starting (Exited (1)).

Diagnosis: docker logs on the container showed the error su: unknown user syslog:adm, indicating the image itself was missing a required system user account — an internal packaging fault in that image, not a configuration mistake.

Resolution: The container was rebuilt using a lightweight alpine:latest image instead. Alpine ships with BusyBox, which includes a built-in syslogd, so busybox syslogd -n could be run directly with no additional package installation needed — avoiding the broken image entirely.

6.5 VPCS Devices Losing IP Configuration on Restart

Issue: Every time the GNS3 project or the host VM was restarted, PC1, PC2, and the Syslog server container lost their previously assigned IP addresses, causing connectivity tests to fail until they were reconfigured.

Diagnosis: This is expected behaviour for VPCS nodes and lightweight containers in GNS3 their IP configuration is held in memory rather than persisted to the node's saved configuration.

Resolution: IP addresses were reassigned at the start of each session (PC1: 192.168.10.10/24, PC2: 192.168.20.10/24, Syslog server: 192.168.10.20/24), while router configurations were preserved permanently across restarts using copy running-config startup-config.

7. Challenges Faced

- Coordinating multiple interdependent security layers (VPN, firewall, routing, AAA) meant that a fault in one layer — such as the missing route for VPN traffic could look like a failure in a completely different layer, making isolation of the root cause slower.
- Running the lab inside GNS3 on a VM introduced infrastructure-level issues (switch state corruption, container image failures) that were unrelated to Cisco configuration knowledge but still had to be diagnosed and worked around under time pressure.

- VPCS and container nodes not persisting their configuration across restarts required re-verifying basic connectivity multiple times throughout the assignment.

8. Conclusion

This assignment successfully implemented a layered, enterprise-grade security architecture across a two-site network in GNS3. An IPsec Site-to-Site VPN now encrypts all inter-site traffic, a Zone-Based Firewall enforces directional traffic policy at the network edge, AAA and SSHv2 secure administrative access to both routers, OSPF is protected with MD5 authentication and constrained with passive interfaces and prefix-list filtering, and Syslog, NTP, and SNMPv3 together provide centralized, time-synchronized, and securely-monitored visibility into the network. Every feature was independently verified using the appropriate Cisco IOS show and debug commands, and end-to-end connectivity was confirmed from the HQ LAN, through the VPN tunnel and firewall, to the Branch LAN. The troubleshooting process particularly diagnosing the switch MAC-learning fault, the duplex mismatch, and the missing VPN route reinforced the importance of working systematically from Layer 1 upward when diagnosing connectivity issues in a secured network.